

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Ratios, rates, proportional relationships, and units

03

10 answers · Rationale-rich · Teach from doc

Q1**90**

The correct answer is 90. It's given that the ratio of x to y is equivalent to the ratio 9 to 5. It follows that $\frac{x}{y} = \frac{9}{5}$. Multiplying each side of this equation by $5y$ yields $\frac{5yx}{y} = \frac{95y}{5}$, or $5x = 9y$. Dividing each side of this equation by 9 yields $\frac{5x}{9} = y$. Substituting 162 for x in this equation yields $\frac{5(162)}{9} = y$, which is equivalent to $\frac{810}{9} = y$, or $90 = y$. Therefore, if the value of x is 162, the value of y is 90.

Q2**D****D.** 80,200

Choice D is correct. The increase in Iceland's population can be found by multiplying the increase in population density, in people per square kilometer, by the area, in square kilometers. It's given that the population density of Iceland was 2.5 people per square kilometer in 1990 and 3.3 people per square kilometer in 2014. The increase in population density can be found by subtracting 2.5 from 3.3, which yields 0.8. It's given that the land area of Iceland was 100,250 square kilometers. Thus, the increase in population is **$0.8(100,250)$** , or 80,200.

Alternate approach: It's given that the population density of Iceland, in people per square kilometer of land area, in 1990 was 2.5. Since the land area of Iceland was 100,250 square kilometers, it follows that the population of Iceland in 1990 was **$2.5(100,250)$** , or 250,625. Similarly, the population of Iceland in 2014 was **$3.3(100,250)$** , or 330,825. The population increase is the difference in the population from 1990 to 2014, or **$330,825 - 250,625$** , which yields 80,200. Therefore, Iceland's population increased by 80,200 from 1990 to 2014.

Choice A is incorrect. This is the population of Iceland in 2014. Choice B is incorrect and may result from dividing 3.3 by 2.5, instead of subtracting 2.5 from 3.3. Choice C is incorrect and may result from dividing the population of Iceland in 1990 by 2.

Q3**B****B.** $26x$

Choice B is correct. It's given that at a particular track meet, the ratio of coaches to athletes is 1 to 26. If one number in a ratio is multiplied by a value, the other number must be multiplied by the same value in order to maintain the same ratio. If there are x coaches at the track meet, multiplying both numbers in the ratio by x yields $1x$ to $26x$, or x to $26x$. Therefore, the expression $26x$ represents the number of athletes at the track meet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**34672**

The correct answer is 34,672. It's given that $1 \text{ mile} = 1,760 \text{ yards}$. It follows that an average speed of 19.700 miles per hour is equivalent to $\frac{19.700 \text{ miles}}{1 \text{ hour}} \times \frac{1,760 \text{ yards}}{1 \text{ mile}}$, or 34,672 yards per hour.

Q5**D****D.** 17,136.0

Choice D is correct. Since $1 \text{ minute} = 60 \text{ seconds}$ and $1 \text{ hour} = 60 \text{ minutes}$, it follows that $1 \text{ hour} = (60)(60)$, or 3,600 seconds. Using this conversion factor, the space station's average speed of 4.76 miles per second is equal to an average speed of

$$\frac{4.76 \text{ miles}}{\text{second}} \times \frac{3,600 \text{ seconds}}{\text{hour}} = \frac{17,136 \text{ miles}}{\text{hour}}, \text{ or } 17,136 \text{ miles per hour.}$$

Choice A is incorrect. This is the space station's average speed in miles per minute. Choice B is incorrect. This is double the space station's average speed in miles per minute, or the number of miles the space station travels on average in 2 minutes. Choice C is incorrect. This is triple the space station's average speed in miles per minute, or the number of miles the space station travels on average in 3 minutes.

Q6**1015**

The correct answer is 1,015. It's given that the first moon orbits the planet every 252 days. Therefore, it takes the first moon 25229, or 7,308, days to orbit the planet 29 times. It's also given that the second moon orbits the planet every 287 days. Therefore, it takes the second moon 28729, or 8,323, days to orbit the planet 29 times. Since it takes the first moon 7,308 days and the second moon 8,323 days, it takes the second moon $8,323 - 7,308$, or 1,015, more days than it takes the first moon to orbit the planet 29 times.

Q7**14/3**

Accept also: 4.666, 4.667

The correct answer is $\frac{14}{3}$. It's given that 3 teaspoons is equivalent to 1 tablespoon. Therefore, 14 teaspoons is equivalent to $14 \text{ teaspoons} \cdot \frac{1 \text{ tablespoon}}{3 \text{ teaspoons}}$, or $\frac{14}{3}$ tablespoons. Note that 14/3, 4.666, and 4.667 are examples of ways to enter a correct answer.

Q8**40260**

The correct answer is 40,260. It's given that 1 furlong = 220 yards and 1 yard = 3 feet. It follows that a distance of 61 furlongs is equivalent to $61 \text{ furlongs} \cdot \frac{220 \text{ yards}}{1 \text{ furlong}} \cdot \frac{3 \text{ feet}}{1 \text{ yard}}$, or 40,260 feet.

Q9**A****A. 3**

Choice A is correct. Since the distance is equal to the amount of time multiplied by a constant, the given equation $d = 55t$ represents a proportional relationship between distance and time in this situation. Since $9k = 3 \cdot 3k$, the time when $t = 9k$ hours is 3 times the time when $t = 3k$ hours. Therefore, the distance traveled after $9k$ hours is 3 times the distance after $3k$ hours.

Choices B and D are incorrect and may result from interpreting the proportional relationship between time and distance as additive rather than multiplicative. Choice C is incorrect and may result from an arithmetic error.

Q10**25**

The correct answer is 25.4. The average speed is the total distance divided by the total time.

The total distance is 11 miles and the total time is 26 minutes. Thus, the average speed is $\frac{11}{26}$ miles per minute. The question asks for the average speed in miles per hour, and there are 60 minutes in an hour; converting miles per minute to miles per hour gives the following:

$$\begin{aligned} \text{Average speed} &= \frac{11 \text{ miles}}{26 \text{ minutes}} \times \frac{60 \text{ minutes}}{1 \text{ hour}} \\ &= \frac{660}{26} \text{ miles per hour} \\ &\approx 25.38 \text{ miles per hour} \end{aligned}$$

Therefore, to the nearest tenth of a mile per hour, the average speed of Paul Revere's ride would have been 25.4 miles per hour. Note that 25.4 and $127/5$ are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

